MILITARY STANDARD

MIL-STD-1320-56
(NAVY)

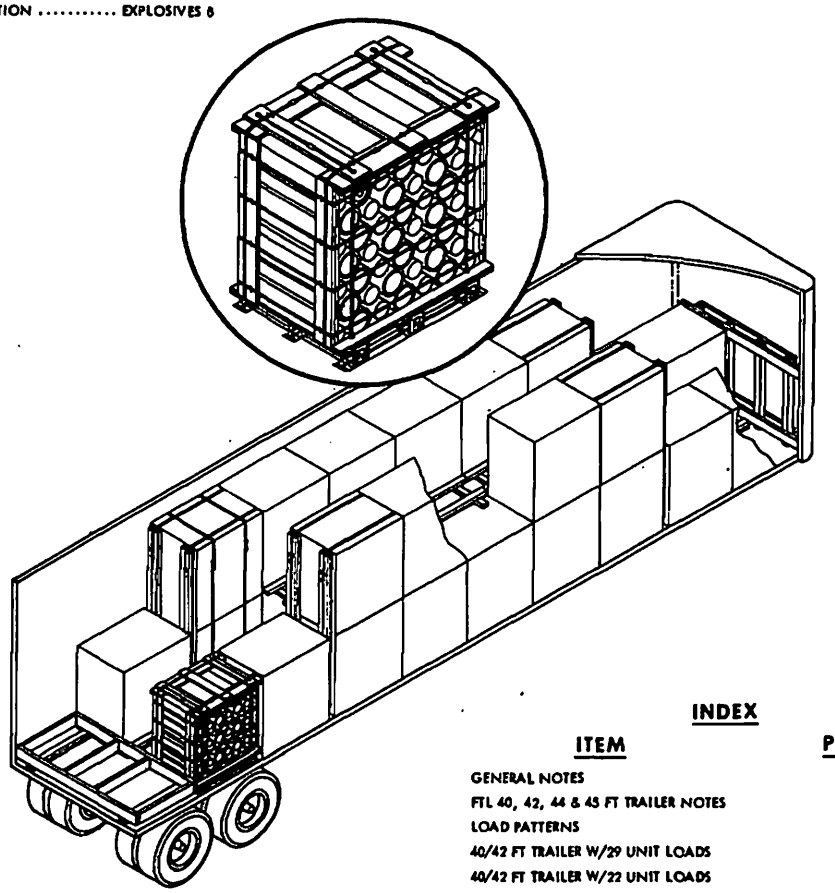
5 APRIL 1976

SUPERSEDING
WR-51/66
3 NOVEMBER 1969

TRUCKLOADING
CHARGE, PROPELLING 5"/38
IN TANK MK 15 MOD 1
DOMESTIC UNIT LOAD

UNIT LOAD DATA

LENGTH	49 1/8
WIDTH	35
HEIGHT	52 1/4
WEIGHT	1706 LBS
DWG NO.	WR-53/757
DOT HAZARD CLASSIFICATION	EXPLOSIVES B



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS-REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDKB-236.

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FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M. S. Lucey NWHC 4/5/76
TECHNICAL DIRECTION AGENT (TDA) DATE

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SEASYSCOM BY DIRECTION DATE

ORIGINATOR L. L. Caringi 4/5/76 DATE
NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING 5"/38 PROPELLING CHARGES IN MK 15 MOD 1 TANKS WHEN PALLETIZED IN ACCORDANCE WITH DOMESTIC UNIT LOAD WR-53/757.
2. THE PROCEDURE AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS. TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. LENGTH OF THE STRUTS, PIECE 8 IN DETAIL L SHALL BE SUFFICIENT TO CAUSE THE BLOCKING ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.
4. THE THICKNESS OF THE SOLID FILL, DETAIL C, SHALL BE SUFFICIENT TO FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.
5. THE LENGTH OF THE STRUTS, PIECE 7, IN DETAIL D SHALL BE SUFFICIENT TO CAUSE THE BLOCKING ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND DETAIL C.
6. THE SWAY BRACE, DETAIL A, IS ASSEMBLED IN PLACE. INSERT SUPPORTS, PIECE 4, AGAINST INBOARD SIDE OF PALLET VERTICAL SUPPORTS. POSITION STRINGERS, PIECE 5, AGAINST PALLET VERTICAL SUPPORT AND NAIL IN PLACE WITH THREE 10D NAILS. WHEN USING SWAY BRACE IN LIEU OF SLEEPERS (PIECE 6), INSTALL BETWEEN THE LOWER LAYER PALLETS IN THE SAME MANNER.
7. TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND LAYER, THE UNIT LOAD SHALL BE UNITIZED (SEE UNITIZING DETAIL PAGE 7) WHERE THE PATTERN CHANGES FROM ONE LAYER HIGH TO TWO LAYERS HIGH. STIFFENER SHALL FACE THE SINGLE LAYER SIDE.
8. WHENEVER THERE ARE UNIT LOADS TWO HIGH OR ONE SIDE OF THE TRAILER AND ONE HIGH ON THE OTHER SIDE (SEE PAGE 4), THE TWO HIGH UNIT LOADS SHALL HAVE TWO ADDITIONAL 1 1/4 X .035 STEEL STRAPS CROSSWISE TO CONTROL LATERAL MOTION IN THE SECOND LAYER. THESE STRAPS ENCIRCLE THE STACKED UNIT LOADS PASSING UNDER THE PALLET DECK OF THE BOTTOM LOAD AND OVER THE TOP OF THE UPPER LOAD. TENSION AND SEAL WITH TWO DOUBLE CRIMPED SEALS.
9. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA. CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVE B" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
11. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
12. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL & LTL 40, 42, 44 & 45 FT TRAILERS (VAN)

WARNING

ALL LOADS THAT HAVE A SECOND LAYER REQUIRE A HIGH CUBE VAN WITH A MINIMUM HEIGHT UNDER ROOF BOWS (AT THE EAVES) OF 108 IN. TO PROVIDE PROPER CLEARANCE FOR THE SECOND LAYER.

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCES HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS. THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES. BEFORE LOADING MORE THAN 22 UNIT LOADS IN A TRAILER DETERMINE THAT THE LIMITATIONS OF THE NOTES IN TABLE II ARE MET.

NOTE

LIGHT WEIGHT TRACTOR-TRAILER – TARE WEIGHT OF TRAILER TANDEM AXLES 6,800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10,500 LBS MAXIMUM.

NOTE

STANDARD WEIGHT TRACTOR-TRAILER – TARE WEIGHT OF TRAILER TANDEM AXLES 7,600 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11,500 LBS MAXIMUM.

TABLE I

TRUCKLOAD WEIGHTS

NO. OF UNIT LOADS	TRUCKLOAD WEIGHT (LBS)
29	49,474
28	47,768
27	46,062
26	44,356
25	42,650
24	40,944
23	39,238
22	37,532

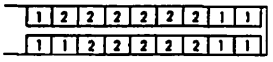
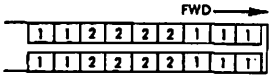
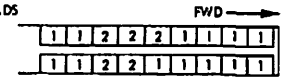
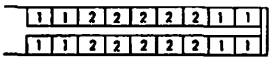
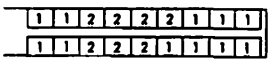
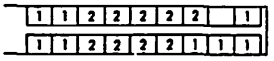
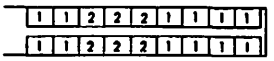
1. TABLE II PROVIDES THE CORRECT LOAD PATTERNS FOR THE LENGTH OF THE TRAILER BEING USED AND THE NUMBER OF UNIT LOADS BEING LOADED.
2. WHEN LOADING TRAILERS WITH A SECOND LAYER, FOLLOW THE PROCEDURES ON PAGE 4, DECREASING THE NUMBER OF UNIT LOADS IN THE SECOND LAYER AS REQUIRED BY THE LOAD PATTERN.
3. WHEN LOADING 22 UNIT LOADS IN A 40 OR 42 FT TRAILER FOLLOW THE PROCEDURES SHOWN ON PAGE 5. WHEN LOADING 24 UNIT LOADS IN A 44 OR 45 FT TRAILER, FOLLOW THE PROCEDURES SHOWN ON PAGE 5, ADJUSTING THE DIMENSIONS OF THE REAR BLOCKING AS REQUIRED BY THE LOAD.
4. WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN TABLE II, FOLLOW THE PROCEDURES SHOWN ON PAGE 6. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

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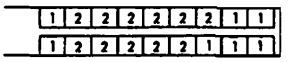
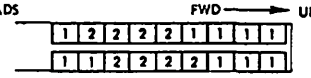
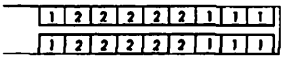
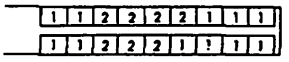
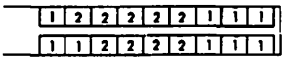
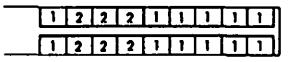
TABLE II

LOAD PATTERNS

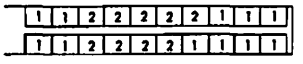
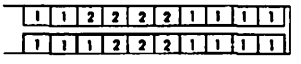
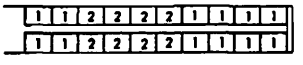
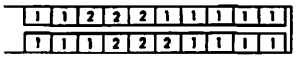
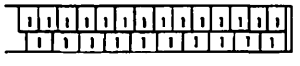
40 FT TRAILERS

NO. OF UNIT LOADS		NO. OF UNIT LOADS		NO. OF UNIT LOADS	
29 ●●●		26 ●		23 ●	
28 ●●●		25 ●		22 ●	
27 ●●		24 ●			

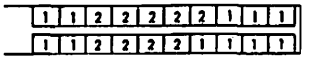
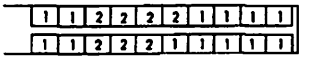
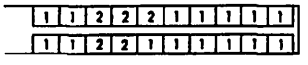
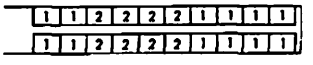
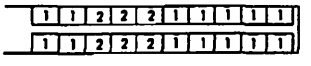
42 FT TRAILERS

NO. OF UNIT LOADS		NO. OF UNIT LOADS		NO. OF UNIT LOADS	
29 ●●●		26 ●		23 ●	
28 ●●●		25 ●		22 ●	
27 ●●		24 ●			

44 FT TRAILERS

NO. OF UNIT LOADS		NO. OF UNIT LOADS		NO. OF UNIT LOADS	
29 ●●●		27 ●●		25 ●	
28 ●●●		26 ●		24 ●	

45 FT TRAILERS

NO. OF UNIT LOADS		NO. OF UNIT LOADS		NO. OF UNIT LOADS	
29 ●●●		27 ●●		25 ●	
28 ●●●		26 ●		24 ●	

●●● AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. ONLY LIGHT WEIGHT TRACTOR-TRAILERS SHALL BE USED.

●● AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THROUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHT OF 73,280 LBS. AUTHORIZED FOR BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND GROSS VEHICLE WEIGHTS OF 80,000 LBS.

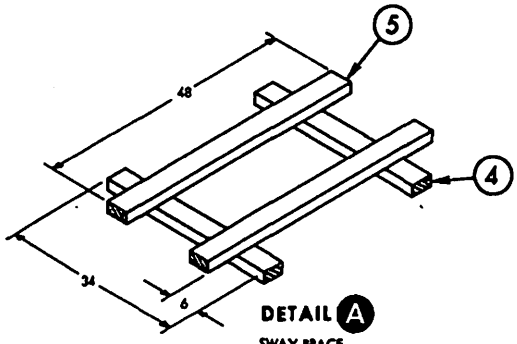
● AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 32,000 LBS ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 73,280 LBS. BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED.

NOTE: LOAD PATTERN SHOWS TRAILER FLOOR PLAN.
NUMBER IN BLOCK INDICATES NUMBER OF UNIT LOADS IN A LAYER.

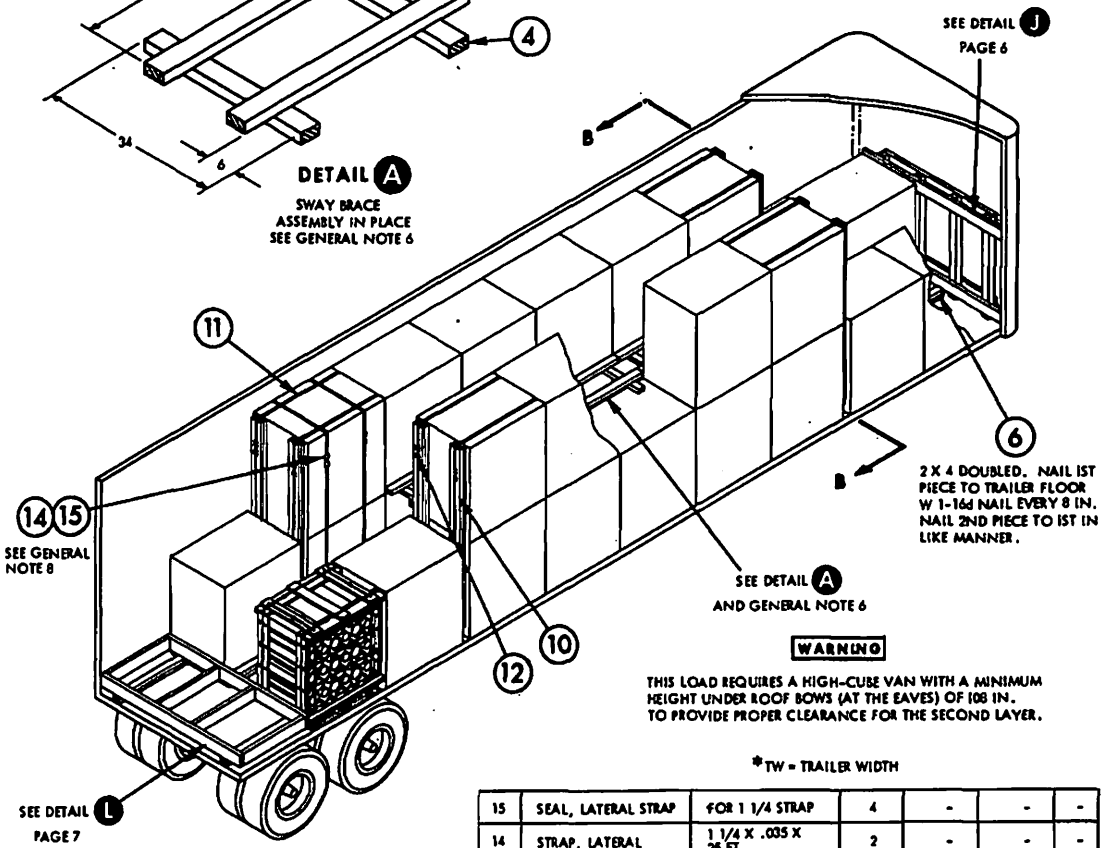
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NOTE

TRUCKLOAD SHOWN IS FOR TRAILERS WITH NAILABLE FLOORS. WHEN A TRAILER HAS A NON-NAILABLE FLOOR, SUBSTITUTE SWAY BRACE, DETAIL A, FOR SLEEPERS PIECE 6. INSTALL BETWEEN PALLETS OF BOTTOM LAYER AS SHOWN FOR UPPER LAYER.



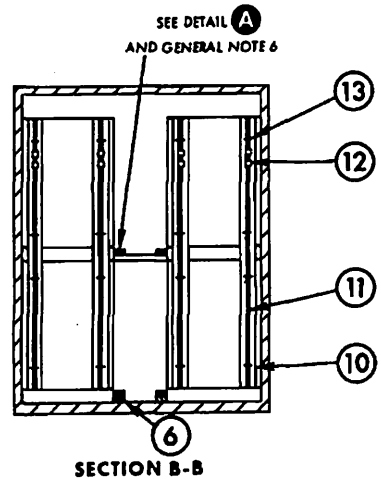
DETAIL A
SWAY BRACE ASSEMBLY IN PLACE SEE GENERAL NOTE 6



2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W 1-16d NAIL EVERY 8 IN. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

WARNING

THIS LOAD REQUIRES A HIGH-CUBE VAN WITH A MINIMUM HEIGHT UNDER ROOF BOWS (AT THE EAVES) OF 108 IN. TO PROVIDE PROPER CLEARANCE FOR THE SECOND LAYER.



SECTION B-B

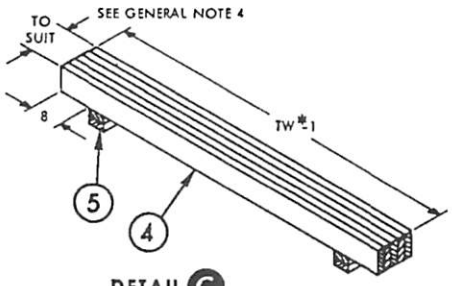
*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
15	SEAL, LATERAL STRAP	FOR 1 1/4 STRAP	4	-	-	-
14	STRAP, LATERAL	1 1/4 X .035 X 25 FT	2	-	-	-
13	STAPLE	FOR 1 1/4 STRAP	32	-	-	-
12	SEAL	FOR 1 1/4 STRAP	16	-	-	-
11	STRAP	1 1/4 X .035 X 25 FT	8	-	-	-
10	STIFFENER	2 X 6 X 99	16	DOUBLED LAMINATED	5	10d
9	SUPPORT	2 X 4 TO SUIT	4	7 9	2/JOINT 4/PIECE	10d 10d
8	STRUT	2 X 6 TO SUIT	4	SEE 7	-	-
7	CROSSMEMBER	2 X 6 X TW*-1	2	8	3/JOINT	16d
6	SLEEPER	2 X 4 X 48	36	SEE FIELD NOTE		
5	STRINGER	2 X 4 X 48	10	4	3/JOINT	10d
4	SUPPORT	2 X 4 X 34	10	SEE 5	-	-
3	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3/JOINT	10d
2	VERTICAL	4 X 4 X 54	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3/JOINT	10d

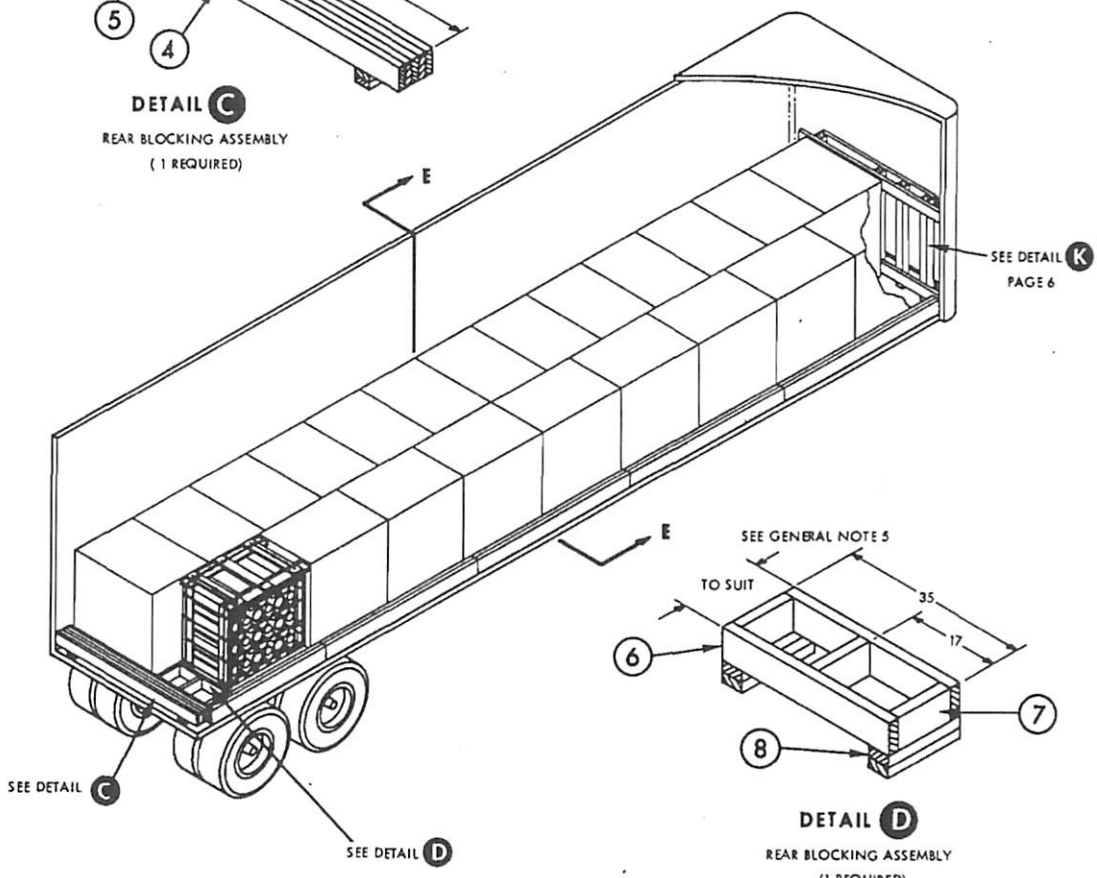
LIST OF MATERIALS & NAILING DATA

40/42 FT TRAILER (VAN) WITH 29 UNIT LOADS

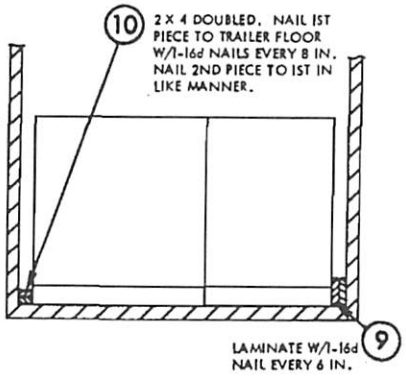
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DETAIL C
REAR BLOCKING ASSEMBLY
(1 REQUIRED)



DETAIL D
REAR BLOCKING ASSEMBLY
(1 REQUIRED)



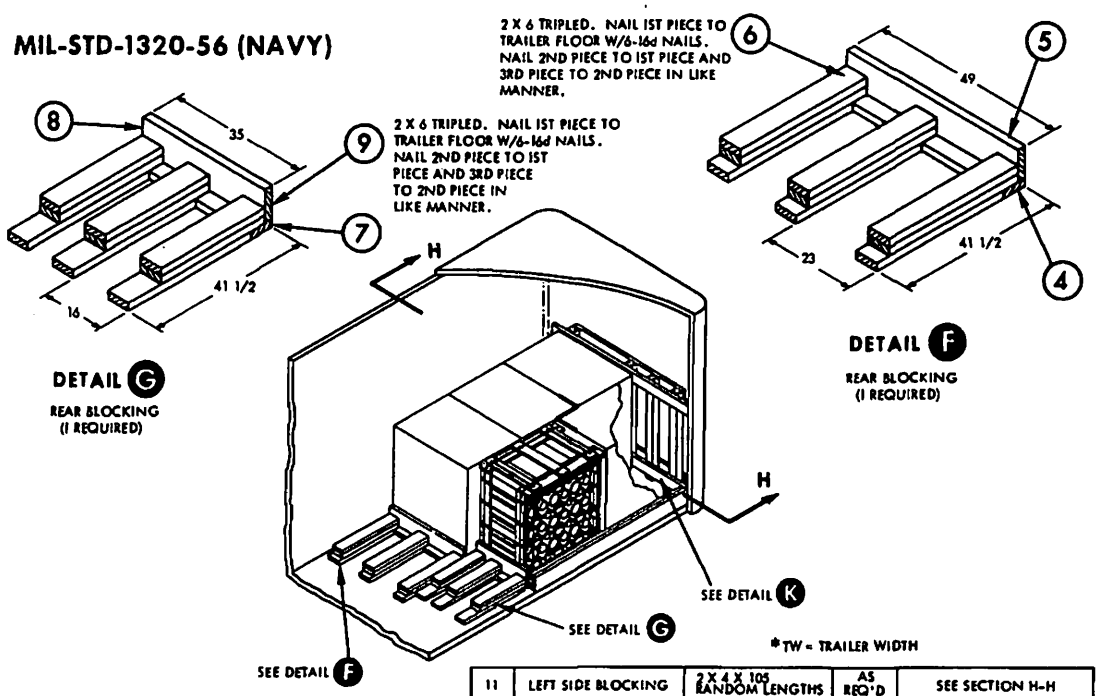
SECTION E-E

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	LEFT SIDE BLOCKING	2 X 4 X 74 FT RANDOM LENGTHS	AS REQ'D	SEE SECTION E-E		
9	RIGHT SIDE BLOCKING	2 X 8 X 76 FT RANDOM LENGTHS	AS REQ'D	SEE SECTION E-E		
8	SUPPORT	2 X 4 TO SUIT	4	6 8	2/JOINT 4/PIECE	10d 10d
7	STRUT	2 X 6 TO SUIT	3	SEE 6	-	-
6	CROSSMEMBER	2 X 6 X 35	2	7	3/JOINT	16d
5	SUPPORT	2 X 4 TO SUIT	4	4 5	2/JOINT 4/PIECE	10d 10d
4	SOLID FILL	THICKNESS TO SUIT X 6 X TW*-1	AS REQ'D	LAMINATE	6/PIECE	TO SUIT
3	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3/JOINT	10d
2	VERTICAL	4 X 4 X 54	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3/JOINT	10d

LIST OF MATERIALS & NAILING DATA

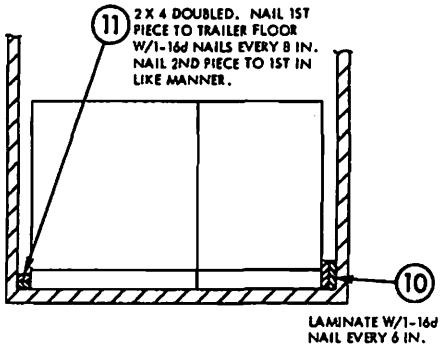
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*TW = TRAILER WIDTH

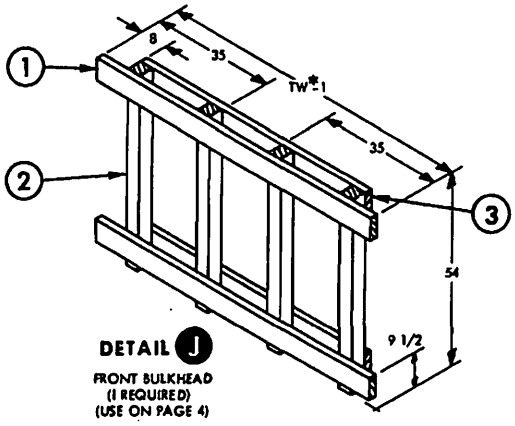
11	LEFT SIDE BLOCKING	2 X 4 X 105 RANDOM LENGTHS	AS REQ'D	SEE SECTION H-H		
10	RIGHT SIDE BLOCKING	2 X 8 X 98 RANDOM LENGTHS	AS REQ'D	SEE SECTION H-H		
9	BACKUP CLEAT	2 X 6 X 30	9	SEE DETAIL G		
8	VERT CROSSMEMBER	2 X 8 X 35	1	SEE 7	-	-
7	BOT CROSSMEMBER	2 X 6 X 35	1	8 TR FLOOR	5 5	16d 16d
6	BACKUP CLEAT	2 X 6 X 36	9	SEE DETAIL F		
5	VERT CROSSMEMBER	2 X 8 X 49	1	SEE 4	-	-
4	BOT CROSSMEMBER	2 X 6 X 49	1	5 TR FLOOR	5 5	16d 16d
3	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3/JOINT	10d
2	VERTICAL	4 X 4 X 54	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3/JOINT	10d
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER OF NAILS	SIZE

LIST OF MATERIALS & NAILING DATA

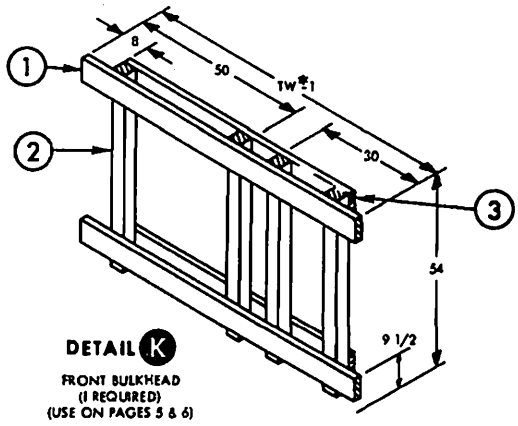


SECTION H-H

TYPICAL LTL



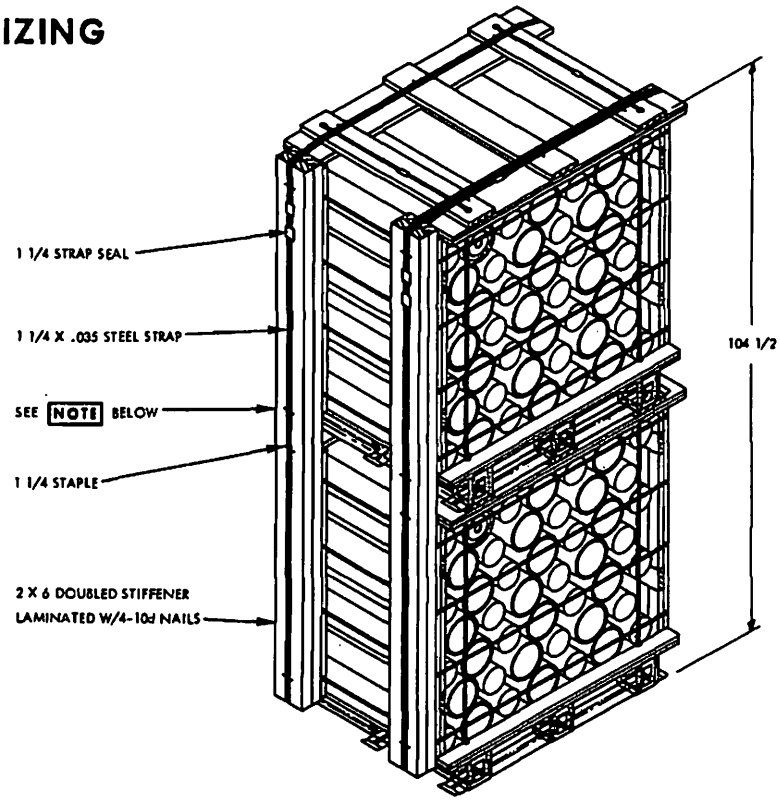
DETAIL J
FRONT BULKHEAD
(1 REQUIRED)
(USE ON PAGE 4)



DETAIL K
FRONT BULKHEAD
(1 REQUIRED)
(USE ON PAGES 5 & 6)

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UNITIZING



WARNING

CHECK HEIGHT OF DOOR OPENING BEFORE UNITIZING. UNITIZING MAY BE DONE INSIDE TRAILER IF DOOR HEIGHT DOES NOT PERMIT PASSAGE OF UNITIZED UNIT LOADS. SEE **WARNING** ON PAGE 4.

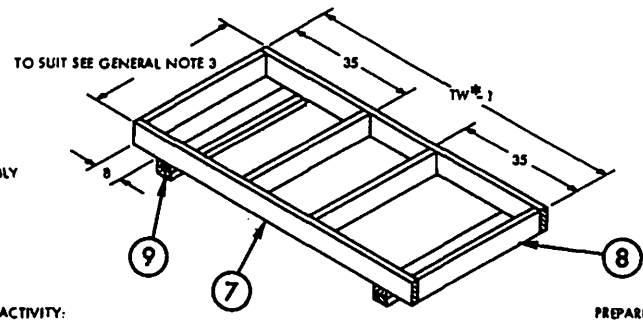
1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD, THE STACKED UNIT LOADS SHALL BE UNITIZED AS SHOWN ABOVE.
2. THE DOUBLED 2 X 6 STIFFENER SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE PALLET OF THE BOTTOM UNIT LOAD.
3. THE 1 1/4 IN. STEEL STRAPS, POSITIONED AS SHOWN, ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET. THE STRAPS HOLD THE STIFFENERS IN PLACE.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 IN. STAPLES.

NOTE

WHEN LOADING TRAILER WITH A SINGLE STACK OF UNIT LOADS, THE STIFFENERS SHALL BE AT BOTH ENDS OF THE UNITIZED LOADS.

DETAIL L

REAR BLOCKING ASSEMBLY (USED ON PAGE 4)



REVIEW ACTIVITY:
NAVY-OS

PREPARING ACTIVITY
NAVY-OS
(PROJECT NO 8140-N328)

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